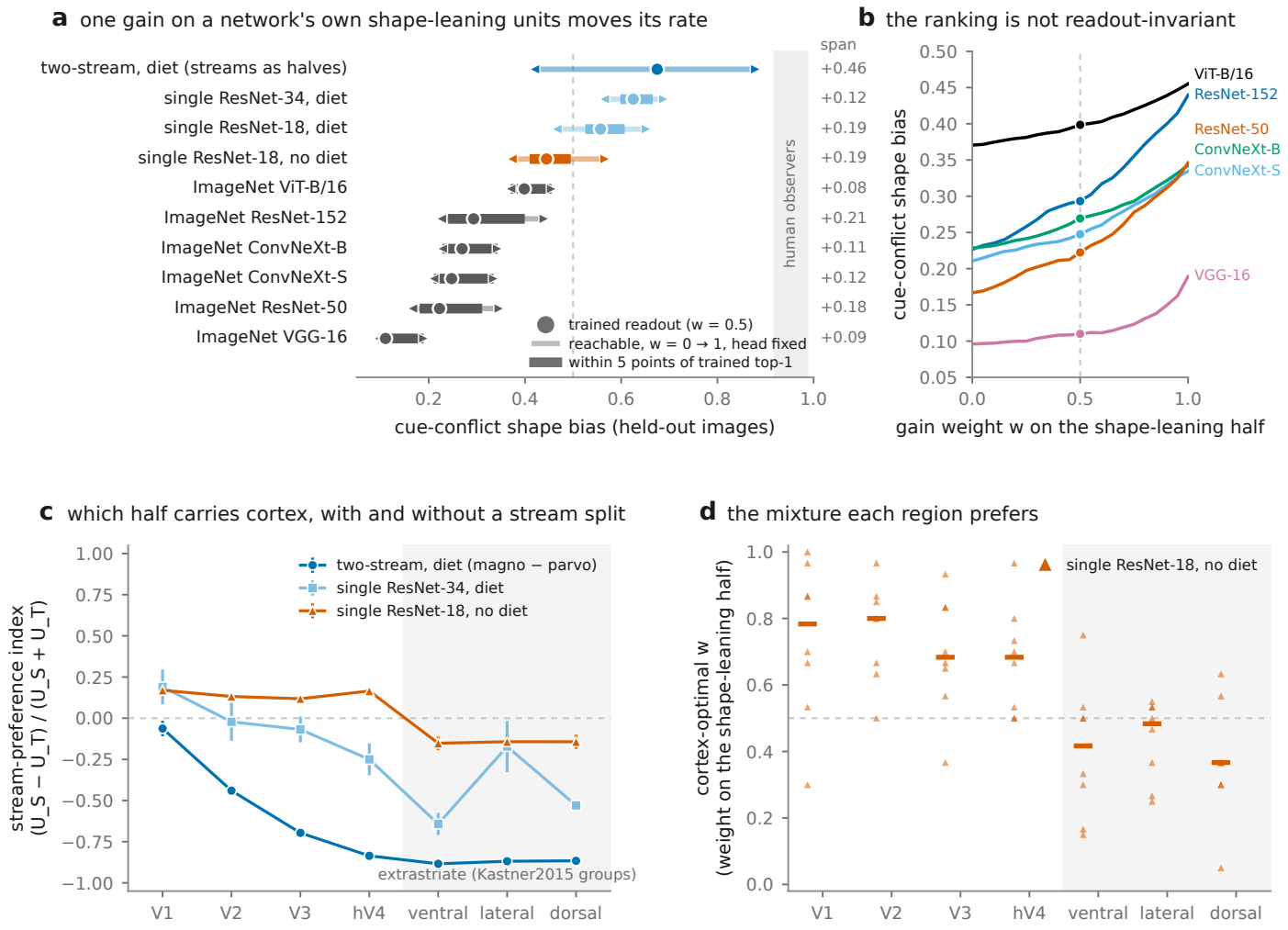
a one gain on a network's own shape-leaning units moves its rate
two-stream, diet (streams as halves)
single ResNet-34, diet
single ResNet-18, diet
single ResNet-18, no diet
ImageNet ViT-B/16
ImageNet ResNet-152
ImageNet ConvNeXt-B
ImageNet ConvNeXt-S
ImageNet ResNet-50
ImageNet VGG-16
span
+0.46
+0.12
+0.19
+0.19
+0.08
+0.21
+0.11
+0.12
+0.18
+0.09
human observers
trained readout (w = 0.5)
reachable, w = 0 → 1, head fixed
within 5 points of trained top-1
0.2
0.4
0.6
0.8
1.0
cue-conflict shape bias (held-out images)
b the ranking is not readout-invariant
cue-conflict shape bias
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.0
0.5
1.0
gain weight w on the shape-leaning half
ViT-B/16
ResNet-152
ResNet-50
ConvNeXt-B
ConvNeXt-S
VGG-16
c which half carries cortex, with and without a stream split
stream-preference index
(U_S − U_T) / (U_S + U_T)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
two-stream, diet (magno − parvo)
single ResNet-34, diet
single ResNet-18, no diet
extrastriate (Kastner2015 groups)
V1
V2
V3
hV4
ventral
lateral
dorsal
d the mixture each region prefers
cortex-optimal w
(weight on the shape-leaning half)
1.0
0.8
0.6
0.4
0.2
0.0
single ResNet-18, no diet
V1
V2
V3
hV4
ventral
lateral
dorsal